

Silver Plating + Faraday's Law

Experiment Packet

Audience: instructors and teaching assistants

Purpose: Bench instructions for instructors and TAs

Session hook: Estimate how many atoms were deposited

Length: 90-minute lab block (~20 min concepts + ~70 min hands-on)

This packet includes

1. Session overview & plan
2. Preparation
3. Materials checklist
4. Experiment procedure
5. Shared safety notes

Generated from the Electrochemistry workshop curriculum. Prefer the latest PDF from the course repository if procedures change mid-week.

Session 3 — Silver Plating and Faraday's Law

Hook: Calculate how many silver atoms you deposit and how thick the film is.

Learning outcomes

- Plate silver onto a metal object with controlled current and time
- Use $Q = It$, $n = Q/F$, and mass/thickness relations
- Discuss assumptions (100% efficiency, uniform coverage)
- Compare ideal Faraday prediction to visual result

Session plan (90 min)

Time	Activity	Reference
0–10 min	Hook: dull copper object — can we count the atoms we add?	lecture.md
10–25 min	Ag^+ reduction and Faraday's law	lecture.md
25–35 min	Estimate plated surface area	experiment.md — Area
35–45 min	Set up cell with ammeter in series	experiment.md — Setup
45–65 min	Plate with timed current readings	experiment.md — Run
65–80 min	Calculate charge, moles, mass, thickness	experiment.md — Calculations
80–90 min	Discuss efficiency and side reactions	lecture.md

Key equations

$$\text{Ag}^+(\text{aq}) + \text{e}^- \rightarrow \text{Ag}(\text{s})$$

$$Q = I \times t$$

$$n_{\text{e}} = Q / F \quad (F = 96,485 \text{ C/mol})$$

$$n_{\text{Ag}} = n_{\text{e}}$$

$$m_{\text{Ag}} = n_{\text{Ag}} \times M_{\text{Ag}} \quad (M_{\text{Ag}} = 107.87 \text{ g/mol})$$

$$d = m_{\text{Ag}} / (\rho_{\text{Ag}} \times A) \quad (\rho_{\text{Ag}} = 10.49 \text{ g/cm}^3)$$

Status

- ☐ AgNO_3 solution prepared (0.01 M)
- ☐ Calculation worksheet ready
- ☐ Pilot plating run completed
- ☐ Waste container labeled for silver

Session 3 — Preparation

Before class (instructor)

- ☐ Prepare 0.01 M AgNO_3 with **distilled/deionized water** (not tap); volume for all groups + 20% extra
- ☐ Pilot plate one object: record I_{avg} , t , appearance
- ☐ Verify 3 V packs deliver stable $\sim 10\text{--}30$ mA in this cell
- ☐ Print calculation worksheets
- ☐ Confirm silver waste disposal with school/facility protocol

AgNO_3 handling

- Wear gloves when preparing
- Work in ventilated area
- Label all bottles clearly
- Store away from organic materials and light if possible

Day-of setup

- ☐ Small volumes pre-measured per station (reduces spills)
- ☐ Goggles enforced at door
- ☐ Waste container visible and labeled
- ☐ Demo calculation on board with example numbers (before students start)

Timing note

Calculations take longer than expected for some groups. Have I_{avg} pre-computed on board as check, or provide a spreadsheet template.

Open questions / notes

- Stock AgNO_3 concentration on hand:
- Volume prepared (mL):
- Pilot I_{avg} (mA):
- Pilot plating time (s):

Session 3 — Materials

Per group

- ☐ 0.01 M AgNO_3 solution (~20–30 mL per group — prepare from stock)
- ☐ Distilled water (for rinsing)
- ☐ Object to plate: copper coin, washer, brass piece
- ☐ 3 V battery pack (preferred) or 9 V with caution
- ☐ Alligator clips
- ☐ Multimeter (current + voltage)
- ☐ Stopwatch
- ☐ Gloves, goggles, forceps
- ☐ Small beaker or jar
- ☐ Ruler (area estimation)

Optional

- ☐ Silver wire or strip (anode — replenishes Ag^+)
- ☐ Balance (for efficiency check)
- ☐ Calipers (diameter measurement)

Instructor-only

- ☐ AgNO_3 stock solution (instructor dilutes to 0.01 M)
- ☐ Dedicated **silver waste** container
- ☐ Spare gloves (staining risk)

Solution prep note

Target: 0.01 M AgNO_3 — instructor prepares before class.

Use distilled/deionized water only (tap chloride can precipitate AgCl).

Batch	Solid AgNO_3	Distilled water
100 mL	0.17 g	to 100 mL
250 mL	0.42 g	to 250 mL
500 mL	0.85 g	to 500 mL

Or dilute 0.1 M stock **1:10** with distilled water. Label date and concentration. ~20–30 mL per group.

Waste

- ☐ All Ag-containing liquids go to labeled waste container

Session 3 — Experiment: Silver Plating + Calculations

Instructors: area pitfalls, ammeter series wiring, I_{avg} — instructor.md.

Silver nitrate solution — concentration and prep

Target working solution: 0.01 M AgNO_3 (instructor prepares before class).

Water: use distilled or deionized water only. Tap water often contains chloride, which precipitates AgCl (cloudy white) and can ruin a dilute silver bath. Rinse plated objects with distilled water too.

Batch	Solid AgNO_3 needed	Distilled water
100 mL of 0.01 M	0.17 g AgNO_3	Fill to 100 mL
250 mL of 0.01 M	0.42 g AgNO_3	Fill to 250 mL
500 mL of 0.01 M	0.85 g AgNO_3	Fill to 500 mL

(Molar mass $\text{AgNO}_3 \approx 169.9 \text{ g/mol} \rightarrow 0.01 \text{ mol/L} \times 169.9 \approx 1.70 \text{ g/L}$.)

If you have a more concentrated stock

Example: dilute **0.1 M** stock **1:10** $\rightarrow 0.01 \text{ M}$

(e.g. 25 mL of 0.1 M + 225 mL **distilled** water = 250 mL of 0.01 M).

Volume per group today

20–30 mL of 0.01 M AgNO_3 — use the minimum that covers the object. Do not prepare large open beakers of AgNO_3 at student benches.

Labeling

0.01 M AgNO_3 – Session 3 – gloves required – silver waste only

Part A — Estimate surface area (25–35 min)

Object options

Copper coin, copper washer, cleaned brass charm, or small flat copper piece.

Methods

Coin (one face):

Measure diameter (d) in cm $\rightarrow (A = \pi (d/2)^2)$.

If both faces plate, use $(A = 2\pi (d/2)^2)$.

Flat washer:

$(A \approx \pi(R_{\text{outer}}^2 - R_{\text{inner}}^2))$ (one side), or $\times 2$ if both sides plate.

Irregular object:

Approximate immersed region as length × width (cm × cm). State the approximation in your notes.

Record

Dimension	Value (cm)	Formula used	Area A (cm ²)

Tip: A typical coin face is ~2–4 cm². If your A is 50 cm², you probably measured wrong — recheck before calculating thickness.

Part B — Setup (35–45 min)

Session 3 - Silver plating with current measurement

multimeter in series
A
mA

Object (-)

3V

0.01 M AgNO₃ (small v)

Cu

silver deposits

Ag anode (+)

$\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag(s)}$ Record I every 30-60 s and total time t

$Q = I \times t \rightarrow n = Q/F \rightarrow m_{\text{Ag}} \rightarrow \text{thickness } d = m/(\rho A)$

Estimate plated area A before starting (coin face, washer ring, etc.)

Figure 1 — Plate the cathode object (-); measure current with the multimeter in series for Faraday calculations.

Power supply

Preferred	Acceptable
3 V battery pack	9 V with short runs and careful current checks

Target current: ~**10–30 mA** (0.010–0.030 A).

Procedure

1. Put on **gloves and goggles**.

- Pour **20–30 mL** of **0.01 M AgNO₃** into a small beaker or jar.
- Clean the cathode object: sand → rinse with distilled water → dry → handle with forceps/gloves only.
- Wire the circuit **in series**:
- Battery (+) → anode (silver wire/strip if available; otherwise an inert conductor — note that Ag⁺ will deplete)
- Anode in solution → solution → object
- Object (–) → multimeter **mA** input → battery (–)
Or: battery → meter → object → solution → anode → battery.
- Set multimeter to **DC mA** (not voltage). The meter must be **in series**, never clipped across the battery terminals alone.
- Brief test: dip electrodes; confirm current is in the 10–30 mA range. If much higher, lift electrodes partially or use 3 V instead of 9 V.

Setup record

Item	Value
Object	
Area A (cm ²)	
AgNO ₃ concentration	0.01 M
Solution volume (mL)	
Anode type	
Supply voltage	
Target plating time (s)	e.g. 300 s
Initial current (mA)	

Part C — Timed plating run (45–65 min)

- Start the stopwatch when a **stable** current is flowing with the object immersed.
- Record current every **30–60 s** for the planned time (recommend **300 s = 5 min**).
- If current drifts, keep logging — you will use the average.
- At stop time: disconnect power first, then remove the object.
- Rinse with distilled water; pat dry. Observe color and uniformity.

Current log

Time (s)	I (mA)	I (A) = mA ÷ 1000	Notes
0			
60			
120			

Time (s)	I (mA)	I (A) = mA ÷ 1000	Notes
180			
240			
300			

Average current:

I_{avg} (mA) = __ → I_{avg} (A) = ____ A

(Example: 20 mA = **0.020 A**)

Part D — Faraday calculations (65–80 min)

Use I_{avg} in amperes and total time t in seconds.

Constants for silver:

Quantity	Value
Faraday constant F	96,485 C/mol
Molar mass Ag	107.87 g/mol
Density ρ_{Ag}	10.49 g/cm ³
Electrons per Ag ⁺	z = 1

Step	Formula	Your calculation	Result
1. Charge	$Q = I_{\text{avg}} \times t$		C
2. Moles e ⁻	$n_e = Q / 96,485$		mol
3. Moles Ag	$n_{\text{Ag}} = n_e$ (because $z = 1$)		mol
4. Mass Ag	$m = n_{\text{Ag}} \times 107.87$		g
5. Volume Ag	$V = m / 10.49$		cm ³
6. Thickness	$d = V / A$		cm
7. Micrometers	$d_{\mu\text{m}} = d_{\text{cm}} \times 10,000$		μm

Worked check example (board / instructor)

$I = 0.020 \text{ A}$, $t = 300 \text{ s}$, $A = 10 \text{ cm}^2 \rightarrow Q = 6.0 \text{ C} \rightarrow n = 6.22 \times 10^{-5} \text{ mol} \rightarrow m \approx 0.0067 \text{ g} \rightarrow d \approx \textbf{0.64 } \mu\text{m}$.

Part E — Compare to reality (80–90 min)

Discussion prompts:

1. Can you **see** a layer ~0.5–1 μm thick?
2. If current dropped, did you use I_{avg} ?

3. What side reactions might steal electrons (e.g. H_2)?
4. Would doubling the time double the thickness in the ideal case?

Optional: efficiency estimate

If a balance (± 0.001 g) is available:

$$\eta = (\text{actual mass gained} / \text{calculated mass}) \times 100\%$$

Safety

- **Gloves and goggles mandatory** — AgNO_3 stains skin and clothing permanently dark brown/black
 - Small volumes only (20–30 mL per group)
 - Dedicated **silver waste** container — **never** pour down the drain
 - Instructor prepares/dilutes AgNO_3 ; students do not handle solid AgNO_3
 - Wash gloves/hands protocol as directed before leaving
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Student calculation sheet (duplicate for handout)

Given: $I_{\text{avg}} = \underline{\hspace{1cm}}$ A, $t = \underline{\hspace{1cm}}$ s, $A = \underline{\hspace{1cm}}$ cm^2

AgNO_3 bath: 0.01 M

$$Q = I \times t = \underline{\hspace{2cm}} \text{ C}$$

$$n_e = Q / 96485 = \underline{\hspace{2cm}} \text{ mol}$$

$$m_{\text{Ag}} = n_e \times 107.87 = \underline{\hspace{2cm}} \text{ g}$$

$$d = m_{\text{Ag}} / (10.49 \times A) = \underline{\hspace{2cm}} \text{ cm} = \underline{\hspace{2cm}} \mu\text{m}$$

Experiment status

- [] 0.01 M AgNO_3 prepared and labeled (volume: $\underline{\hspace{1cm}}$ mL)
- [] Typical current at 3 V recorded in pilot ($\underline{\hspace{1cm}}$ mA)
- [] Worksheet tested with sample numbers
- [] Silver waste protocol confirmed with facility

Syllabus Safety Notes

Apply these across all sessions. Session-specific hazards are listed in each session folder.

General rules

1. **No eating experimental materials.** Fruits, potatoes, and liquids used in experiments are no longer food.
2. **Avoid short circuits.** Batteries can heat up if directly shorted.
3. **Clean surfaces matter.** Electroplating requires clean metal surfaces.
4. **Metal salt waste should be collected.** Copper and silver solutions should not be casually poured down the drain.
5. **Water choice:** tap water is fine for most baths; use **distilled/deionized water for 0.01 M AgNO₃** (Session 3) so chloride does not precipitate silver.

Session-specific hazards

Session	Key hazard	Mitigation
1	Sharp nails/wires	Supervise handling; wash hands after metals
2	Copper solutions	Goggles; no ingestion; collect waste
3	Silver nitrate	Gloves + goggles; stains skin/clothing; small volumes
4	Gas evolution + H ₂ balloon	No saltwater; ammeter in series; instructor-only balloon/flame demo
5	Corrosion demos + electrolytic derusting	Use washing soda or baking soda only — NOT NaCl; disconnect power when not supervised

Hydrogen balloon / pop demo (Session 4, instructor-only)

- Feed **cathode (H₂) gas only** into a **small** balloon — do not mix with O₂
- Inflate modestly; move away from the electrolysis bath and faces
- Long lighter at arm's length; clear area; extinguisher accessible
- **Never** ignite a sealed rigid container or a mixed H₂/O₂ balloon
- Skip entirely if the venue forbids open flame